Research Report: FFmpeg swscaler "Unsupported input" error converting yuv422p10le to gbrpf32le or rgb48le with null TRC (transfer characteristics). Error: fmt:yuv422p10le csp:bt709 prim:bt709 trc:(null) → fmt:gbrpf32le csp:gbr prim:bt709 trc:(null). This happens with Apple ProRes 422 10-bit footage being extracted to EXR. What is the correct minimal FFmpeg filter chain or flag to handle missing TRC metadata for ProRes 10-bit YUV to EXR float conversion? Is this a known FFmpeg bug? What do professional VFX pipelines use?

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Sources: 64 verified

Executive Summary

The minimal FFmpeg fix for the "Unsupported input" error when converting 10-bit ProRes 422 (`yuv422p10le`) with missing Transfer Characteristics (TRC) to a linear float EXR format is to explicitly declare the input TRC using the `-color_trc bt709` flag before the input file. This flag provides the necessary metadata for FFmpeg's `swscale` library to perform the conversion, resolving the `trc:(null)` error with negligible performance impact. The issue is a known FFmpeg problem, documented in bug ticket #11585, where the scaler fails when it lacks a defined transfer function for a YUV-to-RGB conversion.

While the `-color_trc` flag is the most efficient solution for standard Rec.709 footage, it does not support professional log formats like ARRI LogC or Sony S-Log3. For these sources, the correct method is to use the more CPU-intensive `zscale` filter (e.g., `-vf "zscale=t=linear"`), which can be up to 36 times slower than the `swscale` alternative [6]. Professional VFX pipelines and commercial transcoders like DaVinci Resolve and Adobe Media Encoder avoid this error entirely by using robust color management systems that allow users to manually assign the source color space, preventing the conversion from failing due to missing metadata [7, 12]. The root cause of the missing `trc:(null)` metadata is not a single camera but a systemic issue in various NLE and external recorder export workflows [5].

Key Findings

1. Finding: The most direct and minimal FFmpeg fix for converting standard (Rec.709) ProRes with a null TRC to EXR is to use the `-color_trc bt709` flag on the input file.

Confidence: MEDIUM
Sources: Multiple sources

2. Confidence: HIGH

Confidence: HIGH
Sources: Multiple sources

3. Sources: [1, 14, 15]

Confidence: MEDIUM
Sources: Multiple sources

4. Finding: The swscaler error is a documented FFmpeg issue, logged as bug ticket #11585, which describes the identical "Unsupported input" error when a 10-bit source has `trc:(null)`.

Confidence: MEDIUM
Sources: Multiple sources

5. Confidence: HIGH

Confidence: HIGH
Sources: Multiple sources

6. Finding: The `-color_trc` flag does not support log curves; for log-encoded ProRes (e.g., ARRI LogC, Sony S-Log3), the correct method is to use the `zscale` filter to linearize the footage, such as `-vf "zscale=t=linear"`.

Confidence: MEDIUM
Sources: Multiple sources

7. Confidence: HIGH

Confidence: HIGH
Sources: Multiple sources

8. Sources: [4, 21, 23]

Confidence: MEDIUM
Sources: Multiple sources

Detailed Analysis

The Minimal Fix vs. The Robust Fix

The core of the user's problem is that FFmpeg's `swscale` library, responsible for color and pixel format conversions, requires complete information about the input and output formats. The error message `fmt:yuv422p10le csp:bt709 prim:bt709 trc:(null) -> fmt:gbrpf32le csp:gbr prim:bt709 trc:(null)` explicitly shows that while color primaries (`prim:bt709`) and color space (`csp:bt709`) are known, the transfer characteristics (`trc:(null)`) are missing. Without knowing the gamma curve of the input, `swscale` cannot accurately convert the YUV pixel values to a linear RGB format required for EXR and therefore halts with an "Unsupported input" error [1, 2].

1. The Minimal Fix: `-color_trc`
For standard dynamic range footage that uses a Rec.709 gamma curve, the minimal and most performant solution is to provide this missing piece of metadata using an input flag:
`ffmpeg -color_trc bt709 -i input_prores.mov -pix_fmt gbrpf32le output.exr`

This command does not add a processing step; it simply tags the input stream, allowing the highly optimized and multi-threaded `swscale` library to proceed with the conversion [6, 14].

2. The Robust Fix: `zscale` for Log and HDR
The `-color_trc` flag is limited to the enumerated values in FFmpeg's `AVColorTransferCharacteristic`, which includes standards like `bt709` and `smpte2084` (PQ/HDR10) but excludes proprietary camera log curves [4]. For ProRes files containing log-encoded video (e.g., ARRI LogC, Sony S-Log3), the `zscale` filter is required. This filter is a more advanced color conversion engine that can linearize footage from various transfer functions.

- **ARRI LogC:** `ffmpeg -i input.mov -vf "zscale=t=linear:tin=log100" ...`

- **Sony S-Log3:** `ffmpeg -i input.mov -vf "zscale=t=linear:tin=slog3" ...`

For curves not explicitly supported by `zscale`, like Panasonic V-Log, a 3D Look-Up Table (LUT) from the manufacturer is the standard method:

- **Panasonic V-Log:** `ffmpeg -i input.mov -vf "lut3d=path/to/VLog_to_Linear.cube" ...` [23]

Performance Implications

There is a significant performance difference between these two methods. The `-color_trc` flag adds negligible overhead. The `zscale` filter, however, is computationally intensive. For a 4K 10-bit HLG to SDR conversion, `zscale` was measured to be 36 times slower than real-time, largely because it was not multi-threaded in FFmpeg versions prior to 5.1 and relies on slow logarithmic functions [6]. While linearizing a standard gamma curve is less complex, `zscale` remains significantly slower than `swscale`, making `-color_trc` the preferred method whenever applicable.

Professional Pipeline Workflows

Professional VFX pipelines prioritize color accuracy and workflow stability over raw transcode speed. They encounter the same `trc:(null)` issue, but the tools they use are designed to handle it.

- **Commercial Transcoders:** Applications like DaVinci Resolve, Adobe Media Encoder, and Colorfront Transkoder do not fail when metadata is missing. Instead, they either fall back to project settings or provide a user interface for the operator to manually assign the input color space (e.g., "Input Color Space," "Gamma Tag") [7, 12, 13]. This user-driven override prevents the error and ensures correct color interpretation before the conversion to linear EXR.

- **Automated Pipelines (OpenImageIO):** In automated environments, a tool like OpenImageIO's `oiiotool` is often used. It wraps FFmpeg for video reading but provides a more abstract color management layer. The equivalent fix in `oiiotool` is to explicitly declare the source color space before converting to linear, which is a common first step when creating VFX plates [3].

`oiiotool input.mov --iscolorspace Rec709 --tocolorspace linear -d half -o output.exr`

- **Root Cause Prevention:** The `trc:(null)` issue is often a workflow problem. It is frequently traced back to NLEs like DaVinci Resolve or Final Cut Pro where color space tags were not manually specified during export, or to external recorders like Atomos devices with known metadata bugs in older firmware [5, 18, 20]. The best practice is to ensure color metadata is correctly embedded at the source.

Recommendations

1. **For Rec.709 ProRes:** Use the `ffmpeg -color_trc bt709 -i input.mov ...` command. It is the most efficient, minimal, and correct solution for converting standard dynamic range ProRes with missing TRC metadata to EXR.
2. **For Log-Encoded ProRes:** Use the `zscale` video filter to ensure accurate linearization (e.g., `-vf "zscale=t=linear:tin=log100"` for ARRI LogC). Accept the performance decrease as necessary for color fidelity.
3. **For Unsupported Log Curves:** For formats like Panasonic V-Log, use the `lut3d` filter with an official manufacturer 3D LUT to convert to linear space.
4. **Prevent the Issue at the Source:** When exporting from NLEs like DaVinci Resolve, manually set the "Color Space Tag" and "Gamma Tag" under "Advanced Settings." Ensure external recorders have the latest firmware installed to

mitigate known metadata bugs.
5. **For Automated Pipelines:** Employ tools with robust color management like OpenImageIO (`oiiotool`) that allow for explicit declaration of source color spaces (e.g., `--iscolorspace`) as a standard part of the ingestion process.

Limitations & Caveats

- The primary FFmpeg bug ticket referenced (#11585) documents the error for a 10-bit to 8-bit conversion, not 10-bit to float EXR. However, the root cause (`trc:(null)`) and error message are identical, making the ticket highly relevant despite the different output format.

- The performance metric that `zscale` can be "36 times slower" was measured in a specific HLG-to-SDR conversion test. The exact performance penalty for linearizing Rec.709 or LogC footage may vary but will still be substantially higher than using `swscale` .

- While the report identifies the need for a 3D LUT for certain log formats like V-Log, it does not provide the specific LUT files, which users would need to source from the camera manufacturer.

Follow-up Questions

1. What is the performance impact of the `zscale` filter in the latest FFmpeg versions (post-5.1), and have multi-threading improvements been implemented that specifically accelerate the `t=linear` transformation?
2. How does the `prores_metadata` bitstream filter (`-bsf:v prores_metadata=...`) compare in performance and application to using the `-color_trc` input flag for correcting metadata?
3. For a given ProRes file with `trc:(null)` , is there a reliable method to programmatically determine if the intended TRC was Rec.709 or a log curve, short of visual inspection or external metadata files?
4. Which specific versions of DaVinci Resolve, Final Cut Pro, and AtomOS firmware are confirmed to correctly tag all variants of ProRes (SDR, HLG, Log) with the appropriate TRC metadata on export?

Sources

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Methodology

- **Backend**: VERTEX API
- **Model**: gemini-2.5-pro
- **Research Depth**: 3 (follow-up iterations)
- **Research Breadth**: 3 (parallel queries per iteration)
- **Total Sources Evaluated**: 64
- **Quality Threshold**: 40%+ (verified sources only)
- **Duration**: 365.5s
- **Synthesis Method**: AI-assisted cross-referencing and verification

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API Costs

Metric	Value
Input Tokens	28,136
Output Tokens	8,555
Total Tokens	**36,691**
Input Cost	$0.0352
Output Cost	$0.0856
Total Cost	**$0.1207**

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